

ANCHAL GUPTA

Computer Science Undergraduate — Software Engineering

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Professional Overview

Computer Science undergraduate (CGPA: 8.22) with hands-on software engineering experience spanning machine learning, Android development, and embedded IoT systems. Built a CNN-RNN deep learning pipeline that achieved 86.67% deepfake-detection accuracy across 50,000+ video frames, engineered a real-time chat application with authenticated session handling, and delivered a multi-sensor IoT automation system with custom embedded firmware. Comfortable owning problems end-to-end, from data pipelines to production-ready features, with strong foundations in data structures, algorithms, object-oriented programming, and software engineering principles.

Work Experience

Computer Vision and Deep Learning Intern

Jun 2025 – Jul 2025

Centre of Excellence (CoE), IGDTUW — Remote

- Built a modular Python image and video classification pipeline with independently testable preprocessing, training, and evaluation stages.
- Benchmarked model performance across 5,000+ labeled samples and optimized preprocessing and hyperparameters to improve classification reliability.

Projects

Deepfake Detection Model

GitHub

- Designed a hybrid CNN-RNN architecture (Xception + Bi-GRU) to capture spatial and temporal deepfake patterns across 50,000+ video frames.
- Improved temporal prediction stability by tuning sequence length and smoothing strategies, achieving 86.67% classification accuracy.

IoT-Based Smart Office Automation

Embedded Systems

- Developed a smart office automation system integrating RFID, DHT11, MQ gas, Ultrasonic, and LDR sensors using Arduino UNO and ESP8266.
- Implemented custom Embedded C firmware for concurrent sensor polling, automated alerts, and real-time environmental monitoring.

ChatterBox — Real-Time Chat Application

GitHub

- Built a real-time chat application with secure user authentication, centralized state management, and persistent session handling.
- Designed a responsive HTML/CSS interface ensuring smooth communication under varying network conditions.

Education

Indira Gandhi Delhi Technical University for Women — B.Tech, Computer Science

2024 – Present

CGPA: 8.22 — New Delhi, India

Kendriya Vidyalaya — Class XII (CBSE)

2024

Percentage: 89.2%

Kendriya Vidyalaya — Class X (CBSE)

2022

Percentage: 94%

Technical Skills

- **Languages:** Python, C++, JavaScript, C, PHP, Embedded C
- **Development:** REST APIs, HTML, CSS, Kotlin (Jetpack Compose), SQL, MySQL, Firebase
- **Tools:** Git, GitHub, VS Code, Android Studio, Google Colab, Kaggle, Arduino IDE, ThingSpeak, Keil

Soft Skills

- **Leadership & Ownership:** Led a 20-member cross-functional R&D team and managed end-to-end execution of technical initiatives.
- **Communication & Collaboration:** Presented technical findings to stakeholders and coordinated multiple community-driven events.
- **Problem Solving:** Resolved machine learning optimization and embedded hardware integration challenges under tight deadlines.

Achievements & Leadership

- Secured **2nd Prize** at TechFest AAKRITI Ideathon (TIIPS Noida, 2024), **Top 5** at IEEE IGDTUW Hackathon (2025), and **Top 50** among 2,500+ participants at CodeSlayer Hackathon, NIT Delhi (2025).
- **Team Lead, R&D — E-Cell IGDTUW:** Led a 20-member research team conducting company analysis and preparing stakeholder reports.
- **Creative Team Head, Aarohan — IGDTUW:** Directed event planning, venue management, and execution for the university cultural fest.